

CX 4220/CSE 6220 High Performance Computing Spring 2023 Midterm Exam I

February 22, 2023

NAME:

GT Username:

Note:

- The exam has 5 questions for a total of 25 points. Please make sure that you have all the questions.
- Clearly state any assumptions you make in the solution of a problem.
- If the space provided is not sufficient, you may use the additional sheet provided at the end. Do mention in your solution that you have used the additional sheet.
- Run-time for n element 1) parallel prefix or 2) computing the sum operation, for any binary associative operator, is given by:

Computation time: $\left(\frac{n}{p} + \log p\right)$

Communication time: $((\tau + \mu) \log p)$

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Question	Max-Points	Points
1	5	
2	5	
3	5	
4	5	
5	5	
Total	25	

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1. (5 points) The run-times for solving a problem are estimated in clock cycles as follows:

$$\text{Serial run-time} \approx 4n$$

$$\text{Parallel run-time} \approx 8\frac{n}{p} + 1000 \log p$$

- (a) What is the maximum speedup possible as a function of p (consider $n \gg p$)? (2 points)
- (b) For $n = 2.5$ million, what is the largest number of processors that can be used while ensuring efficiency of $\frac{1}{3}$? Approximate calculation is sufficient; for example, $2^{10} \approx 1000$. (1.5 points)
- (c) What is the smallest problem size that must be solved to use 64 processors and achieve efficiency of at least $\frac{1}{3}$? (1.5 points)

2. (5 points) The best sequential and parallel algorithms for solving a problem have the following run-times:

$$\begin{aligned}T(n, 1) &= \Theta(n) \\T(n, p) &= \Theta\left(\frac{n}{p} \log n + \log p\right), \quad p \leq n\end{aligned}$$

- (a) What is the maximum efficiency, as a function of n , achievable by this parallel algorithm? (2 points)
- (b) What is the maximum number of processors (as a function of n) that can be used while still achieving the maximum possible efficiency? (3 points)

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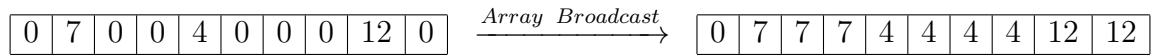
3. (5 points) Given $x_0, x_1, x_2, \dots, x_{n-1}$ and $y_0, y_1, y_2, \dots, y_{n-1}$, compute $s_0, s_1, s_2, \dots, s_{n-1}$, where

$$\begin{aligned} s_0 &= y_0 \\ s_i &= x_i \times s_{i-1} + y_i \quad \text{for } 1 \leq i \leq n-1 \end{aligned}$$

Design a parallel algorithm for solving this problem and compute its run-time.

(**Hint:** Consider using parallel prefix.)

4. (5 points) Consider an array A that contains both zero and non-zero elements. The *Array Broadcast* operation copies each non-zero element to replace the zero elements to its right until the next non-zero element occurs. For example,



Design a parallel algorithm for *Array Broadcast* and specify its run-time.

(**Hint:** Consider using parallel prefix.)

5. (5 points) Consider n elements distributed on p processors such that processor P_i initially has n_i elements. Let $n = \sum_{i=0}^{p-1} n_i$ and $n_{max} = \max_{i=0}^{p-1} n_i$. We want to redistribute the elements such that each processor has approximately the same number of elements. Design an algorithm to solve this problem and analyze its running time. Your algorithm should run in $O(n_{max} \log p)$ computation time and $O((\tau + \mu n_{max}) \log p)$ communication time.

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Extra Page for Solutions
